

Cover sheet for submission of work for assessment

UNIT DETAILS

Unit name	Data Science Principles	Class day/time	Monday	Office use only	
Unit code	COS10022	Assignment no.	1	Due date	6 Oct at 17:59
Name of lecturer/teacher	Dr. Mai Duc Tho				
Tutor/marker's name	Dr. Mai Duc Tho				Faculty or school date stamp

STUDENT

ZFamily Name(s)	Given Name(s)	Student ID Number(s)
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Swinburne University of Technology Hawthorn Campus Dept. of Computing Technologies

COS10022 Data Science Principles Assignment 1 - Semester 1, 2024

Assessment Title: Predictive Model Creation and Evaluation

Assessment Weighting: 20%

Due Date: Sunday, 24th March 2024 at 11.59 pm (AEDT)

Assessable Item:

- One (1) piece of a written report no more than 10-page long with the signed Assignment Cover Sheet.
- The submitted report must be checked by Turnitin, and the similarity from **not the template part** should be less than 12%.

The submitted report should answer all questions listed in the assignment task section in sequence.

You must include a digitally signed Assignment Cover Sheet with your submission.

● Data Preparation (10%)

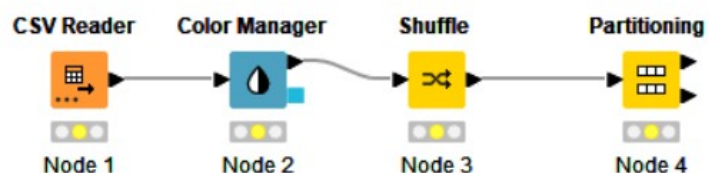
■ Data preparation for training and test set:

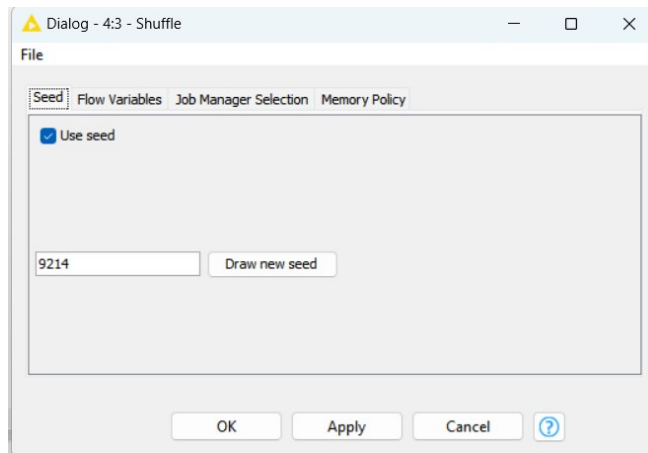
1. For data preparation, we used four essential nodes:

CSV Reader, Color Manager, Shuffle, and Partitioning. The CSV Reader was employed to load data from the file

"Fish_Species.csv," containing 150 tuples and 6 attributes:

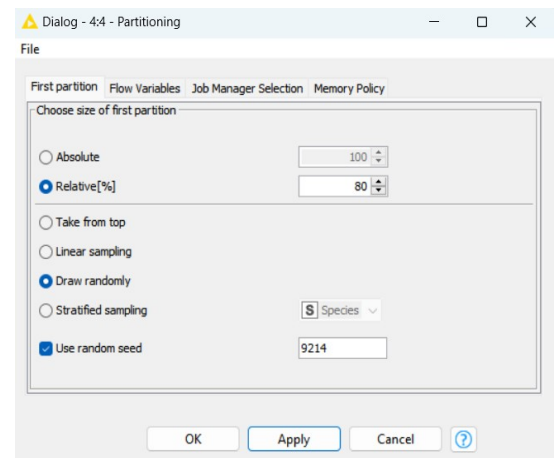
"Weight_of_Fish_in_Gram," "Diagonal_Length_in_cm," "Vertical_Length_in_cm," "Cross_Length_in_cm," "Height_in_cm," and "Diagonal_Width_in_cm." The Color Manager was used to assign colors to different species, aiding data visualization and information recall. To eliminate potential bias during training, the Shuffle node was applied with a **seed value of 9214** for reproducibility. Lastly, the Partitioning node **split the dataset into 80% for training and 20% for testing** using the "draw randomly" method, again with the same seed value of 9214. The training set contains 120 tuples, while the test set includes 30 tuples.





2. By applying the Shuffle Node, the data is randomized to reduce any potential sampling bias that could affect the algorithm's performance. The dataset is shuffled using the seed value of 9214 to ensure consistent and reproducible results. Shuffling the data before partitioning ensures that both the training and test sets contain a diverse representation of the data, which helps improve the model's ability to generalize.

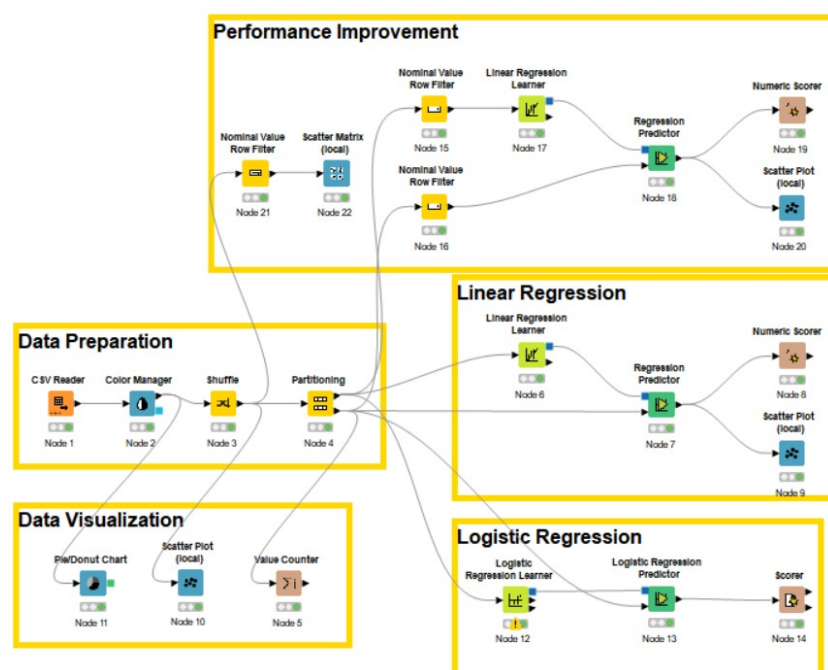
3. Following the data shuffling process, the Partitioning Node is utilized to split the dataset into 80% for training and 20% for testing. This ensures that the training and test sets are distinct from each other. The partitioning is done using the "draw randomly" method with a seed value of 9214 to maintain consistency and reproducibility. The training set is then used to build and train the model, which is later evaluated on the test set to predict its output values.



■ Answering the assignment questions:

- 1) Submit the workflow of Assignment 1 via Assignment 1.1. [2.5 marks]

Answer:

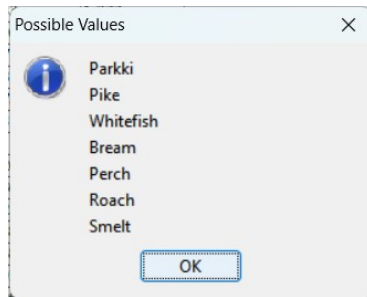


- 2) How many tuples are included in the training set? [2.5 marks]

Answer: **120 tuples**

Explain: The total number of tuples is 150 and Training set is 80% so: $150 \times 80\% = 120$ tuples

- 3) How many species are included in the test set? [2.5 marks]



Answer: There are **7 species** in the test set including "Parkki", "Pike", "Whitefish", "Bream", "Perch", "Roach", and "Smelt".

- 4) Do species "Whitefish" and "Smelt" have the same number of tuples included in the test set? [2.5 marks]

Answer: **NO**. By using the Value Counter Node, we can see that the number of tuples for two species "Whitefish" and "Smelt" are not the same. "Smelt" has 3 tuples while "Whitefish" has only 2 tuples

File Edit Hilite Navigation View		
Table "default" - Rows: 7 Spec - Column: 1		
Row ID		count
Bream		3
Parkki		2
Perch		15
Pike		3
Roach		2
Smelt		3
Whitefish		2

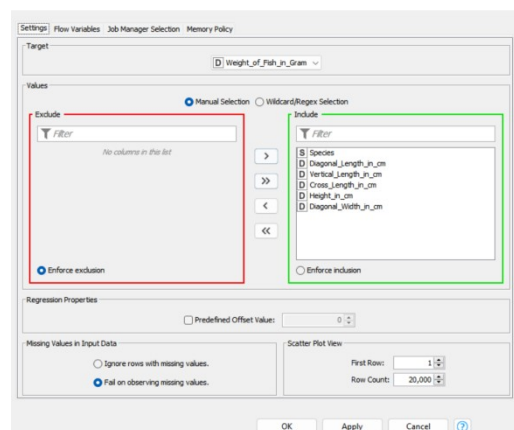
● Linear Regression (40%)

■ Linear Regression Model building :

To successfully build a Linear Regression Model for predicting the weight of the fish, I divided it to 3 small steps:

1. Building the Linear Regression Model:

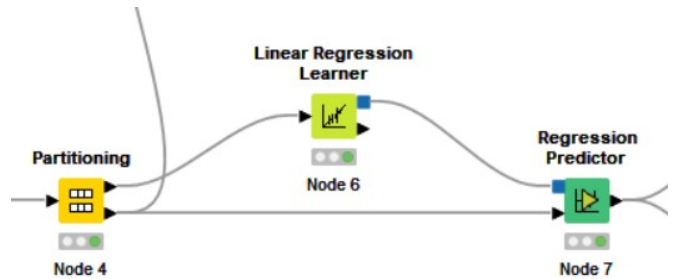
- Add the **Linear Regression Learner** node:
 - Connect the output from the Partitioning node's training set to this node
 - Set the **target column** as "Weight_of_Fish_in_Gram"
 - Make use of all possible attributes for training the model
 - Execute the node to train the linear regression model



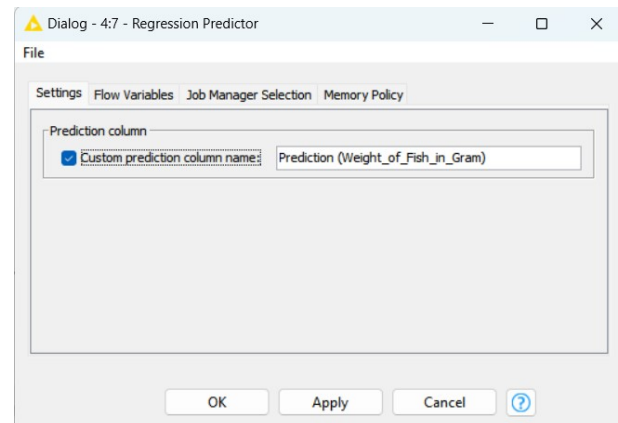
2. Predicting on Test Set:

- Add the **Linear Regression Predictor** node (this node is used for data predicting) :

- Connect the testing data partition and the trained model in the previous step to this node.



- Configuration: Select the Prediction Column and the model will predict the weight of the fish on the test data.

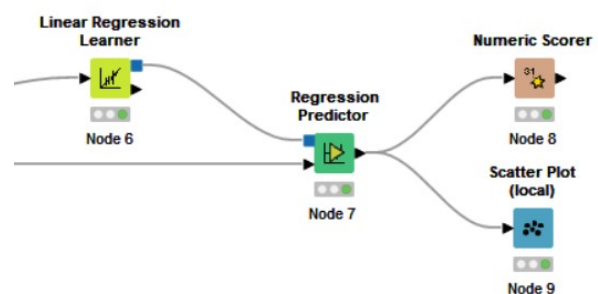


- Execute the node to get predictions for "Weight_of_Fish_in_Gram"

Row ID	S Species	D Weight...	D Diagon...	D Vertical...	D Cross...	D Height...	D Diagon...	D Predict...
Row86	Perch	172.3	23.5	21.5	25	6.275	3.725	223.858
Row88	Perch	147.7	24	22	25.5	6.375	3.825	241.494
Row55	Perch	41.2	15	13.8	16	3.824	2.432	-44.635
Row25	Parkii	947	41	38	46.5	17.623	6.37	884.575
Row60	Bream	199.9	23	21.2	25.8	10.346	3.664	285.996
Row66	Perch	53.4	16.2	15	17.2	4.592	2.632	9.857
Row81	Perch	112.8	22	20	23.5	5.522	3.995	171.317
Row35	Pike	143.8	22	20.5	24.3	6.634	3.548	189.167
Row56	Bream	149.4	20	18.4	22.4	8.893	3.293	180.767
Row125	Roach	456.9	42.5	40	45.5	7.28	4.322	706.681
Row149	Smelt	20.6	15	13.8	16.2	2.932	1.879	-82.428
Row127	Roach	539.1	43	40.1	45.8	7.786	5.13	738.437
Row64	Perch	34.4	13.7	12.5	14.7	3.528	1.999	-94.856
Row93	Perch	300.1	27.3	25.3	28.7	8.323	5.137	401.336
Row99	Perch	319.2	30	27.8	31.6	7.616	4.772	439.224
Row46	Whitefish	270.4	26	23.6	28.7	8.38	4.248	308.579
Row79	Perch	131.1	22	20	23.5	6.11	3.525	176.28
Row68	Perch	99.5	18	16.2	19.2	5.222	3.222	99.605
Row51	Whitefish	1,000.4	40	37.3	43.5	12.354	6.525	790.559
Row29	Pike	87.4	19.8	18.2	22.2	5.617	3.175	92.645
Row110	Perch	851.8	40	36.9	42.3	11.929	7.106	805.476
Row116	Perch	1,100.6	43	40.1	45.5	12.512	7.417	907.733
Row44	Pike	273.7	27	25	30.6	8.568	4.774	343.586
Row114	Perch	1,099.8	42	39	44.6	12.8	6.868	871.34
Row52	Bream	56.6	14.7	13.5	16.5	6.848	2.326	0.606
Row137	Smelt	9.5	10.5	10	11.6	1.972	1.16	-210.632
Row75	Perch	127.6	21	19	22.5	5.692	3.667	142.564
Row73	Perch	109.5	21	19	22.5	5.692	3.555	140.298
Row141	Smelt	5.8	11.3	10.8	12.6	1.978	1.285	-191.643
Row6	Parkii	449.9	30	27.6	35.1	14.005	4.844	507.066

3. Evaluating the Model and Visualizing the Results:

- Add **Numeric Scorer** and **Scatter Plot** nodes:
 - These nodes are used for evaluation and visualization by comparing predicted weights with the actual weights in the test set
 - Connect the output of the **Linear Regression Predictor** node to these nodes



■ Answering the assignment questions:

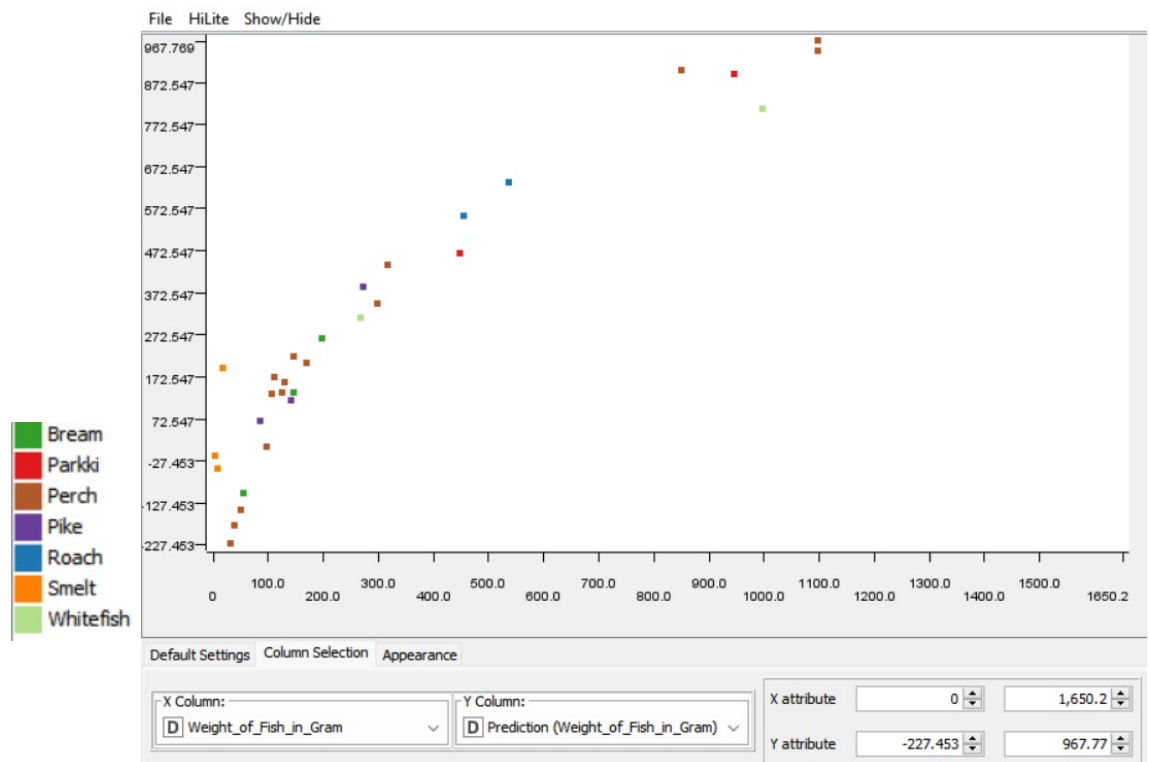
- 1) What is the R^2 value of your test result? [5 marks]

Answer: 0.89

Explain: We utilized the **Numeric Scorer** node to evaluate the performance of our linear regression model, which generated several statistical measures including R^2 , MSE, MAE, RMSE, MSD, MAPE, and Adjusted R^2 .

Statistics - ...	
File	
R^2 :	0.89
Mean absolute error:	87.876
Mean squared error:	12,506.045
Root mean squared error:	111.83
Mean signed difference:	-21.106
Mean absolute percentage error:	1.453
Adjusted R^2 :	0.89

- 2) Give the screenshot of the scatter plot result of your test output using "Weight_of_Fish_in_Gram" on the x-axis and the prediction value on the y-axis. Assign different colours to the data points based on the "species." [15 marks]



- 3) Which species has the heaviest predicted weight in your test result? [5 marks]

Answer: Perch

Explain: According to the predicted data table we can clearly see that "Perch" has the highest predicted number of 967.77

Row ID	S Species	D Weight...	D Diagon...	D Vertical...	D Cross...	D Height...	D Diagon...	D ...
Row116	Perch	1,100.6	43	40.1	45.5	12.512	7.417	967.77
Row114	Perch	1,099.8	42	39	44.6	12.8	8.868	944.783
Row110	Perch	851.8	40	36.9	42.3	11.929	7.106	899.496
Row25	Parkki	947	41	38	46.5	17.623	6.37	889.731
Row51	Whitefish	1,000.4	40	37.3	43.5	12.354	6.525	807.225
Row127	Roach	539.1	43	40.1	45.8	7.786	5.13	631.047
Row125	Roach	456.9	42.5	40	45.5	7.28	4.322	553.657
Row6	Parkki	449.9	30	27.6	35.1	14.005	4.844	462.272
Row99	Perch	319.2	30	27.8	31.6	7.616	4.772	435.7
Row44	Pike	273.7	27	25	30.6	8.568	4.774	385.405
Row93	Perch	300.1	27.3	25.2	28.7	8.323	5.137	343.267
Row46	Whitefish	270.4	26	23.6	28.7	8.38	4.248	309.584
Row50	Bream	199.9	23	21.2	25.8	10.346	3.664	260.238
Row68	Perch	147.7	24	22	25.5	6.375	3.825	219.415
Row86	Perch	172.3	23.5	21.5	25	6.275	3.725	203.552
Row149	Smelt	20.6	15	13.8	16.2	2.932	1.879	190.437
Row81	Perch	112.8	22	20	23.5	5.522	3.995	168.546
Row79	Perch	131.1	22	20	23.5	6.11	3.525	157.471
Row75	Perch	127.6	21	19	22.5	5.692	3.667	133.128
Row56	Bream	149.4	20	18.4	22.4	8.893	3.293	131.623
Row73	Perch	109.5	21	19	22.5	5.692	3.555	130.936
Row35	Pike	143.8	22	20.5	24.3	6.634	3.548	114.016
Row29	Pike	87.4	19.8	18.2	22.2	5.617	3.175	64.52
Row58	Perch	99.5	18	16.2	19.2	5.222	3.322	5.018
Row141	Smelt	5.8	11.3	10.8	12.6	1.978	1.285	-17.366
Row137	Smelt	9.5	10.5	10	11.6	1.972	1.16	-49.031
Row52	Bream	56.6	14.7	13.5	16.5	6.848	2.326	-106.397
Row66	Perch	53.4	16.2	15	17.2	4.592	2.632	-147.402
Row65	Perch	41.2	15	13.8	16	3.824	2.432	-182.96
Row64	Perch	34.4	13.7	12.5	14.7	3.528	1.999	-227.453

- 4) How many prediction results are infeasible in your test result? [5 marks]

Answer: 6

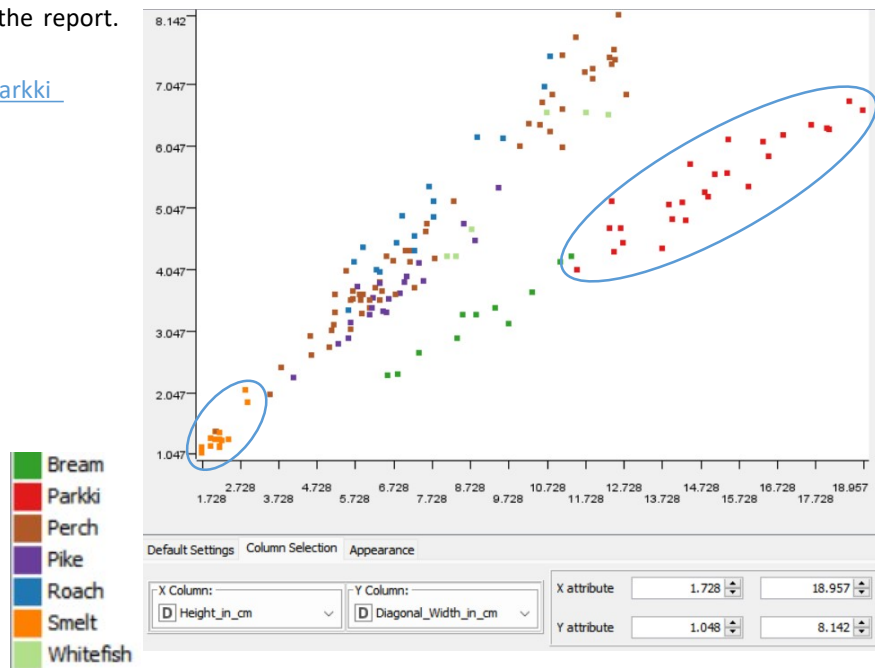
Explain: In the right-bottom of the table, there are 6 predicted weights which are negative number. These are definitely infeasible results because "Weight_of_Fish_in_Gram" must be a positive number

Row ID	S Species	D Weight...	D Diagon...	D Vertical...	D Cross...	D Height...	D Diagon...	D Pre...
Row116	Perch	1,100.6	43	40.1	45.5	12.512	7.417	967.77
Row114	Perch	1,099.8	42	39	44.6	12.8	6.868	944.783
Row110	Perch	851.8	40	36.9	42.3	11.929	7.196	899.496
Row25	Parkki	947	41	38	46.5	17.623	6.37	889.731
Row51	Whitefish	1,000.4	40	37.3	43.5	12.354	6.525	807.225
Row127	Roach	539.1	43	40.1	45.8	7.786	5.13	631.047
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Row60	Bream	199.9	23	21.2	25.8	10.346	3.664	260.238
Row88	Perch	147.7	24	22	25.5	6.375	3.625	219.415
Row86	Perch	172.3	23.5	21.5	25	6.275	3.725	203.552
Row149	Smelt	20.6	15	13.8	16.2	2.932	1.879	190.437
Row81	Perch	112.8	22	20	23.5	5.522	3.995	168.546
Row79	Perch	131.1	22	20	23.5	6.11	3.525	157.471
Row75	Perch	127.6	21	19	22.5	5.692	3.667	133.128
Row56	Bream	149.4	20	18.4	22.4	8.893	3.283	131.623
Row73	Perch	109.5	21	19	22.5	5.692	3.555	130.936
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Row137	Smelt	9.5	10.5	10	11.6	1.972	1.16	-49.031
Row52	Bream	56.6	14.7	13.5	16.5	6.848	2.326	-106.397
Row66	Perch	53.4	16.2	15	17.2	4.592	2.632	-147.402
Row65	Perch	41.2	15	13.8	16	3.824	2.432	-182.96
Row64	Perch	34.4	13.7	12.5	14.7	3.528	1.999	-227.453

- 5) Looking at your source data before splitting them, which species can be easily separated from others if looking at the "Height_in_cm" and "Diagonal_Width_in_cm" attributes? Post your visualisation result on data observation in the report.

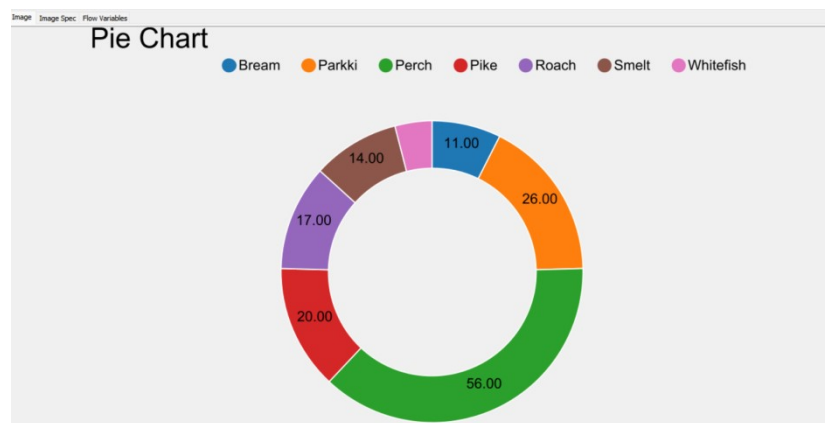
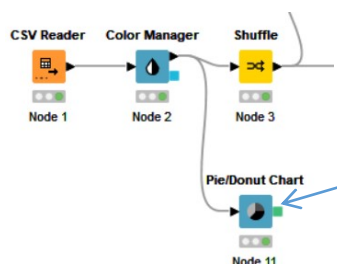
[5 marks]

Answer: Smelt and Parkki



- 6) Draw a doughnut chart of the original input data with 0.55 as the doughnut hole ratio before splitting it into training and test sets. Use different colours for each species and show the percentage of data in the pie chart. [5 marks]

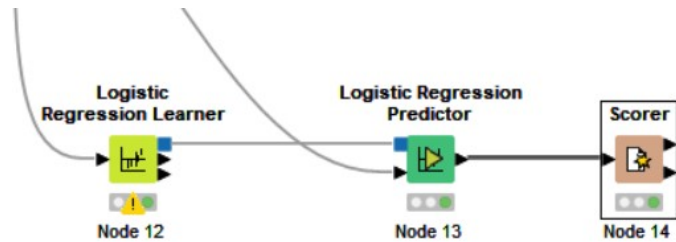
On this task, I included a **Pie/Donut Chart node** to draw a doughnut chart of the original input data from the CSV file with 0.55 hole ratio with various colors for each species:



● Logistic Regression (40%)

■ Logistic regression model building :

Logistic regression model includes a **Logistic Regression Learner node**, a **Logistic Regression Predictor node** and **Scorer node**. I used Smelt as a reference category and the termination points of epochs and epsilon are limited to 10.000 and 0.00001 respectively, use "LineSearch" as the learning rate strategy. Use **9214** as the seed in the logistic regression node



■ Answering the assignment questions:

- 1) Which species have/has no "True Positive (TP)" case in the prediction result? [5 marks]

Answer: Whitefish

Table "default" - Rows: 8 Spec - Columns: 11 Pr

Row ID	TruePo...	FalsePo...
Parkki	2	0
Pike	2	1
Whitefish	0	0
Bream	3	0
Perch	13	3
Roach	2	0
Smelt	3	1
Overall	?	?

- 2) For the species with no TP case, which species will be misplaced? [5 marks]

Answer: Perch

Row99	Perch	319.2	30	27.8	31.6	7.616	4.772	Perch
Row46	Whitefish	270.4	26	23.6	28.7	8.38	4.248	Perch
Row79	Perch	131.1	22	20	23.5	6.11	3.525	Perch
Row68	Perch	99.5	18	16.2	19.2	5.222	3.322	Perch
Row51	Whitefish	1,000.4	40	37.3	43.5	12.354	6.525	Perch
Row29	Pike	87.4	19.8	18.2	22.2	5.617	3.175	Pike

- 3) What is the overall accuracy of the prediction result? [5 marks]

Answer: 0.833

Row ID	TruePo...	FalsePo...	TrueNe...	FalseNe...	D Recall	D Precision	D Sensitivity	D Specificity	D F-meas...	D Accuracy
Parkki	2	0	28	0	1	1	1	1	?	?
Pike	2	1	26	1	0.667	0.667	0.667	0.963	0.667	?
Whitefish	0	0	28	2	?	?	?	?	?	?
Bream	3	0	27	0	1	1	1	1	?	?
Perch	13	3	12	2	0.867	0.812	0.867	0.8	0.839	?
Roach	2	0	28	0	1	1	1	1	?	?
Smelt	3	1	26	0	1	0.75	1	0.963	0.857	?
Overall	?	?	?	?	?	?	?	?	?	0.833

- 4) List all species names with 100% correctly classified test results. [15 marks]

Answer: Parkki, Bream, Roach

Explain: Based on the below formula to calculate each species name and 3 above species have the results equal 1 (100% accuracy).

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

Row ID	TruePo...	FalsePo...	TrueNe...	FalseNe...
Perch	13	3	12	2
Pike	2	1	26	1
Smelt	3	1	26	0
Parkki	2	0	28	0
Whitefish	0	0	28	2
Bream	3	0	27	0
Roach	2	0	28	0

- 5) Which species has a 33.33% chance of being misplaced into another species in the test result? [5 marks]

Answer: Pike

Explain: $FNR = 1 - Sensitivity/Recall$
 $= 1 - 0.667$
 $= 0.333 (= 33.33\%)$

Row ID	TruePo...	FalsePo...	TrueNe...	FalseNe...	D Recall
Perch	13	3	12	2	0.867
Pike	2	1	26	1	0.667
Smelt	3	1	26	0	1
Parkki	2	0	28	0	1
Whitefish	0	0	28	2	0
Bream	3	0	27	0	1
Roach	2	0	28	0	1
Overall	?	?	?	?	?

- 6) In the test result, what percentage of the species "Perch" is misplaced into others? [5 marks]

Answer: 13.3 %

Explain: The percentage of the species "Perch" which are misplaced into others is the percentage of species "Perch" is classified incorrectly to another species name. We have the same formula:

$$\text{FNR} = 1 - \text{Sensitivity/Recall}$$

$$= 1 - 0.867$$

$$= 0.133 (= 13.3 \%)$$

Row ID	TruePo...	Fals...	TrueNe...	FalseN...	Recall
Perch	13	3	12	2	0.867
Pike	2	1	26	1	0.667
Smelt	3	1	26	0	1
Parkki	2	0	28	0	1
Whitefish	0	0	28	2	0
Bream	3	0	27	0	1
Roach	2	0	28	0	1
Overall	?	?	?	?	?

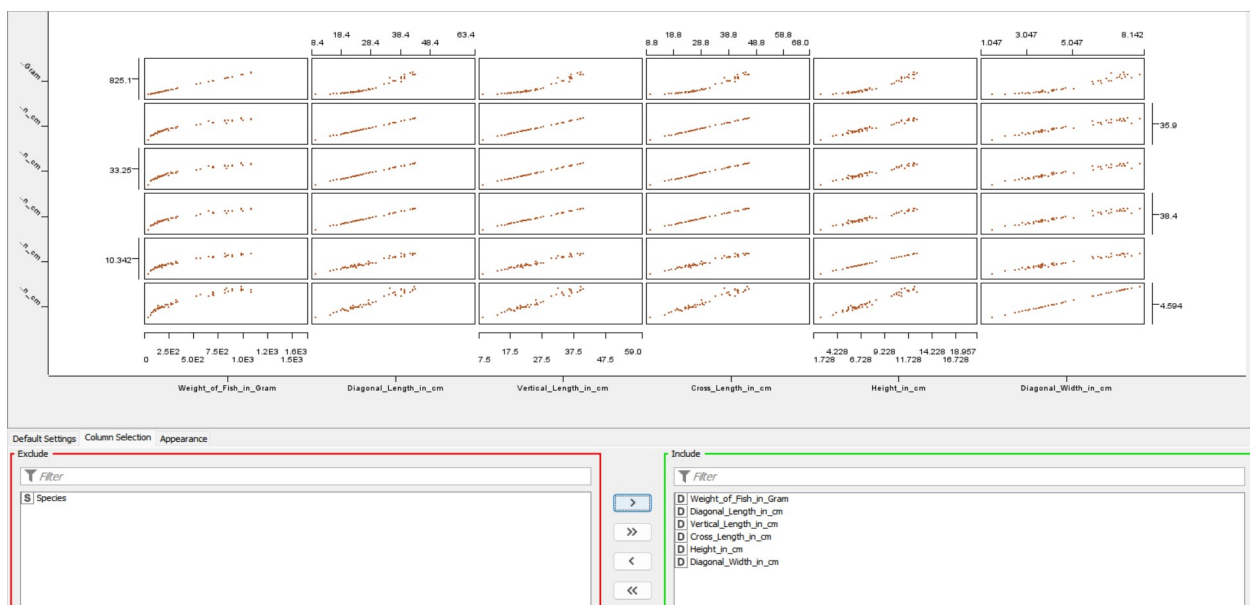
● Performance Improvement (10%) :

■ Build a new linear regression mode:

The goal of this task is to build a more accurate linear regression model, focusing specifically on predicting the weight of the fish species "Perch." By improving the model's accuracy, we aim to achieve a higher R^2 value compared to the previous model.

To concentrate solely on the "Perch" species, two **Nominal Value Row Filter** nodes is used to filter the data, ensuring that only the relevant tuples containing "Perch" are included in both the training and test sets. This filtering process occurs after partitioning the shuffled input data, ensuring that the resulting subsets remain part of the original training and test sets used in earlier stages of the assignment.

I am limited to using three attributes from the available five, which necessitates the removal of two attributes. To facilitate this decision, the **Scatter Matrix (local)** node is employed, enabling us to visually assess the relationships between the input attributes and choose the most suitable ones for the model.



■ Answering the assignment questions:

- 1) Give the reasons for each eliminated attribute and why they are not selected as the input. [5 marks]

Answer: Based on the data from the Scatter Matrix node, two attributes are: "Diagonal Width in cm" and "Weight of Fish in Gram"

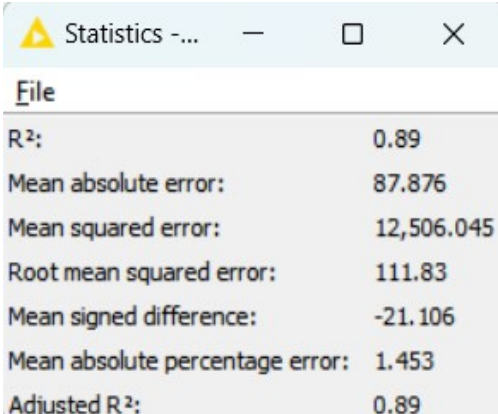
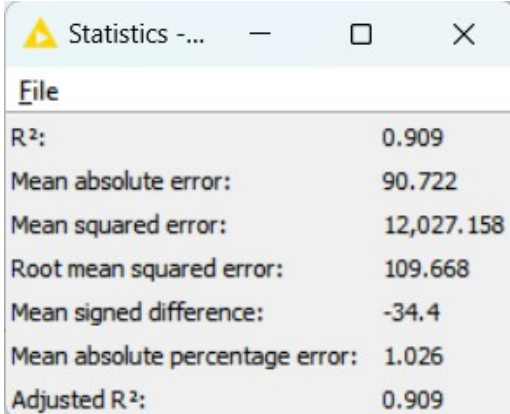
Reasons:

1. Eliminate **"Diagonal_Width_in_cm"** because it shows a weaker linear relationship with **"Weight_of_Fish_in_Gram"** compared to other variables. Including it may introduce noise rather than valuable information, and it could also contribute to multicollinearity, as it is highly correlated with other length measurements.
2. Eliminate **"Height_in_cm"** because it exhibits a weaker linear relationship with **"Weight_of_Fish_in_Gram"** compared to other attributes. Its correlation with weight is less pronounced, which can introduce noise and reduce the model's predictive accuracy. Additionally, keeping this variable may lead to multicollinearity issues, as it is likely correlated with other length measurements.

These eliminations will improve the overall accuracy and stability of the regression model.

- 2) List the R^2 of your test result and compare it with the one in question 2. Reveal both R^2 values obtained in question 2 and in question 4. If you can improve the model, you get the mark. [5 marks]

Answer: _____

First model	Improved model																												
 <table> <tr><td>R^2:</td><td>0.89</td></tr> <tr><td>Mean absolute error:</td><td>87.876</td></tr> <tr><td>Mean squared error:</td><td>12,506.045</td></tr> <tr><td>Root mean squared error:</td><td>111.83</td></tr> <tr><td>Mean signed difference:</td><td>-21.106</td></tr> <tr><td>Mean absolute percentage error:</td><td>1.453</td></tr> <tr><td>Adjusted R^2:</td><td>0.89</td></tr> </table>	R^2 :	0.89	Mean absolute error:	87.876	Mean squared error:	12,506.045	Root mean squared error:	111.83	Mean signed difference:	-21.106	Mean absolute percentage error:	1.453	Adjusted R^2 :	0.89	 <table> <tr><td>R^2:</td><td>0.909</td></tr> <tr><td>Mean absolute error:</td><td>90.722</td></tr> <tr><td>Mean squared error:</td><td>12,027.158</td></tr> <tr><td>Root mean squared error:</td><td>109.668</td></tr> <tr><td>Mean signed difference:</td><td>-34.4</td></tr> <tr><td>Mean absolute percentage error:</td><td>1.026</td></tr> <tr><td>Adjusted R^2:</td><td>0.909</td></tr> </table>	R^2 :	0.909	Mean absolute error:	90.722	Mean squared error:	12,027.158	Root mean squared error:	109.668	Mean signed difference:	-34.4	Mean absolute percentage error:	1.026	Adjusted R^2 :	0.909
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A higher R^2 value indicates improved accuracy in prediction results, and in this case, the model's accuracy has increased by 0.1. This enhancement demonstrates that removing unsuitable attributes to reduce the dimensionality of the input data contributes significantly to the model's overall predictive accuracy.

● Conclusion :

The assignment successfully achieved its goals of analyzing data and developing predictive and classification models. A linear regression model was created to predict fish weight, while a logistic regression model was used for classifying fish species. By refining the linear regression model to focus solely on one species and utilizing only three relevant attributes, the accuracy of predictions improved significantly through the removal of two unsuitable attributes. This process not only enhanced the model's performance but also provided me with valuable skills in data preparation, model analysis, evaluation, and optimization.